MIL-S-2143B(SHIPS)
7 June 1967
SUPERSEDING
MIL-S-2143A(SHIPS)
5 February 1958

MILITARY SPECIFICATION

# SINKS, SERVICE, NAVAL SHIPBOARD

## 1. SCOPE

1.1 This specification covers service (slop) sinks for shipboard use.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-766 - Steel Plates, Sheets, and Strip____Corrosion Resisting.
TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

MILITARY

MIL-F-1183 - Fittings, Tube, Cast Bronze, Silver Brazing.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 11).
MIL-C-15328 - Primer, Pretreatment (Formula No. 117 For Metals).
MIL-C-15726 - Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings.
MIL-B-16444 - Bronze, Hydraulic (Ounce Metal): Castings.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

## 3. REQUIREMENTS

3.1 Material. - Material shall be as specified hereinafter.

3.1.1 Sink. - The sinks shall be made from copper nickel alloy in accordance with MIL-C-15726, or corrosion resisting steel in accordance with QQ-S-766, as specified (see 6.1).



Source: https://assist.dla.mil -- Downloaded: 20261O-16T14:26Z
Check the source to verify that this is the current version before use.

3.1.2 Fittings. - Fittings shall be made of bronze in accordance with MIL-B-16444.

3.2 Design and construction. - Service sinks shall be fabricated from sheet metal 0.0781 ± 0.005 inch thick, with sink compartment, back and rim welded together, complete, to form a one piece unit. The corners shall be rounded or coved on a minimum radius of 1 inch or a maximum radius of 2 inches and at the intersection shall be spherical in shape. The bottom of the sink shall be sloped toward the drain at the center and creased diagonally to provide the necessary pitch for complete drainage. The drain outlet shall be sufficiently countersunk to permit the strainer to set not less than 1/16 inch below the bottom surface of the sink. An open roll rim shall be provided integral with the front and sides of the sink compartment. The back of the sink shall be box type in construction with 2-inch wide flanges on top and sides. Holes for installation of sink faucets shall be provided in the back only when specified and then in the sizes specified (see 6.1). For the purpose of mounting, two 2 by 1/8-inch minimum thick plates of nickel-copper or same material as sink, to each of which are welded two 1/4-inch diameter studs properly spaced to provide secure fastening on the sink supports as specified hereinafter, shall be tack welded to the underside of the sink compartment. Corrosion resistant steel hexagon headnuts and washers for these studs shall be provided.

3.2.1 Dimensions and weight. - The service sink shall be as shown on figure 1. The sink, complete with trap and supports, shall weight approximately 70 pounds.

3.2.2 Strainer. - Each service sink shall be provided with a bronze flanged drain fitting incorporating a flat strainer plate and a threaded spud for connection to the trap inlet. The joint between the sink and the trap shall be sealed by a rubber gasket. The strainer plate shall be removable and shall be retained by at least 2 flat head screws. The strainer plate, screws and fitting shall be chromium plated.

3.2.3 Trap. - Each sink shall be fitted with a 3-inch iron pipe size (i.p.s.) cast bronze P-trap. The trap shall have a brass 1-1/2 inch screw clean-out plug below the water surface. Depth of water seal shall be not less than 2 inches. The trap inlet shall be flanged to provide a seating surface for the rubber gasket of the drain fitting. The trap shall be provided with an adjustable support to the deck. The trap outlet connection shall be a 3-inch i.p.s. silver brazed joint in accordance with MIL-F-1183.

3.2.4 Supports. - Each service sink shall be provided with two steel angle brackets for attachments to bulkhead. The brackets shall be fabricated from 1-1/4 by 1/8 inch steel angles with a 1 by 1/8 inch diagonal steel brace and shall be provided with slotted holes on the horizontal or bearing arm to receive the mounting studs on the sink. The vertical arm shall be provided with one 7/16-inch slotted hole and one 7/16-inch drilled hole for attachment to bulkhead. The brackets shall be finished with two coats of gray enamel conforming to class 2 of MIL-E-15090 applied to thoroughly cleaned metal treated with material conforming to either MIL-C-15328 or TT-C-490. The sink back shall be fitted with a 2 by 1/8 inch horizontal cross bar welded flush with the external edges for a support. Two bulkhead clips 2 by 1/8 by 6 inches long, offset 1/8 inch, and drilled for two 1/4-inch bolts for attachment to bulkhead shall be provided (see figure 1).

3.2.5 Finish. - The inside of the sink shall be finished smooth with a dull polish. Exposed surface of fittings shall be polished. All joints and seams shall be arc welded, the parts fused together in smooth watertight seams. A commercial number 4 satin finish shall be applied to all surfaces inside the sink, the exposed surfaces of the back, ends, and the rolled rims. The sides and bottom of the sink shall have a commercial number 3 brush finish. All exposed welds shall be ground smooth to blend with the finished surfaces. Traps shall be smooth finished externally.

3.3 Workmanship. - Workmanship shall be in accordance with commercial manufacturing practice.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. - All sinks offered for delivery at one time shall be considered a lot for purposes of examination.



Source: https://assist.dla.mil -- Downloaded: 20261O-16T14:26Z
Check the source to verify that this is the currentversion before use.

4.2.1 Sampling for examination. - A random sample of sinks shall be selected from each lot offered for examination in accordance with MIL-STD-105 at inspection level III for lots of 40 and under, inspection level II for lots of 41 to 300, and inspection level I for lots of 301 and over. The Acceptable Quality Level (A.Q.L.) shall be equal to 1.5 percent defective.

4.3 Examination. - Each of the sample sinks selected in accordance with 4.2.1 shall be examined to verify compliance with the requirements of this specification regarding design, dimensions, quality of construction and materials used. Any sink in the sample containing one or more defects shall not be offered for delivery, and if the number of defective sinks in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall not be offered for delivery.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Sinks. -

5.1.1.1 Preservation and packaging. - Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Level A. - Sinks shall be individually packaged in accordance with method III of MIL-P-116. Contact preservative is not required.

5.2.1.1.2 Level C. - Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2.1.2 Packing. -

5.2.1.2.1 Level A. - Sinks packaged as specified in 5.2.1.1 shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

| Specification | Classification |
|---|---|
| PPP-B-576 | Style A, class 2 |
| PPP-B-585 | Style 2, class 2 |
| PPP-B-591 | Class II, style A or B |
| PPP-B-601 | Overseas, style A or B |
| PPP-B-621 | Class 2, style 2 |
| PPP-B-636 | Type SF, Weather-Resistant class |



Source: https://assist.dla.mil -- Downloaded: 20261O-16T14:26Z
Check the source to verify that this is the current version before use.

5.2.1.2.2 Level C. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.2.1.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Material to be used (see 3.1.1).
(c) Size of holes, when required, for back of sink (see 3.2).
(d) Levels of preservation, packaging, packing and marking required if other than as specified in 5.1 (see 5.2).

Preparing activity:
Navy - SH
(Project 4510-N031Sh)



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

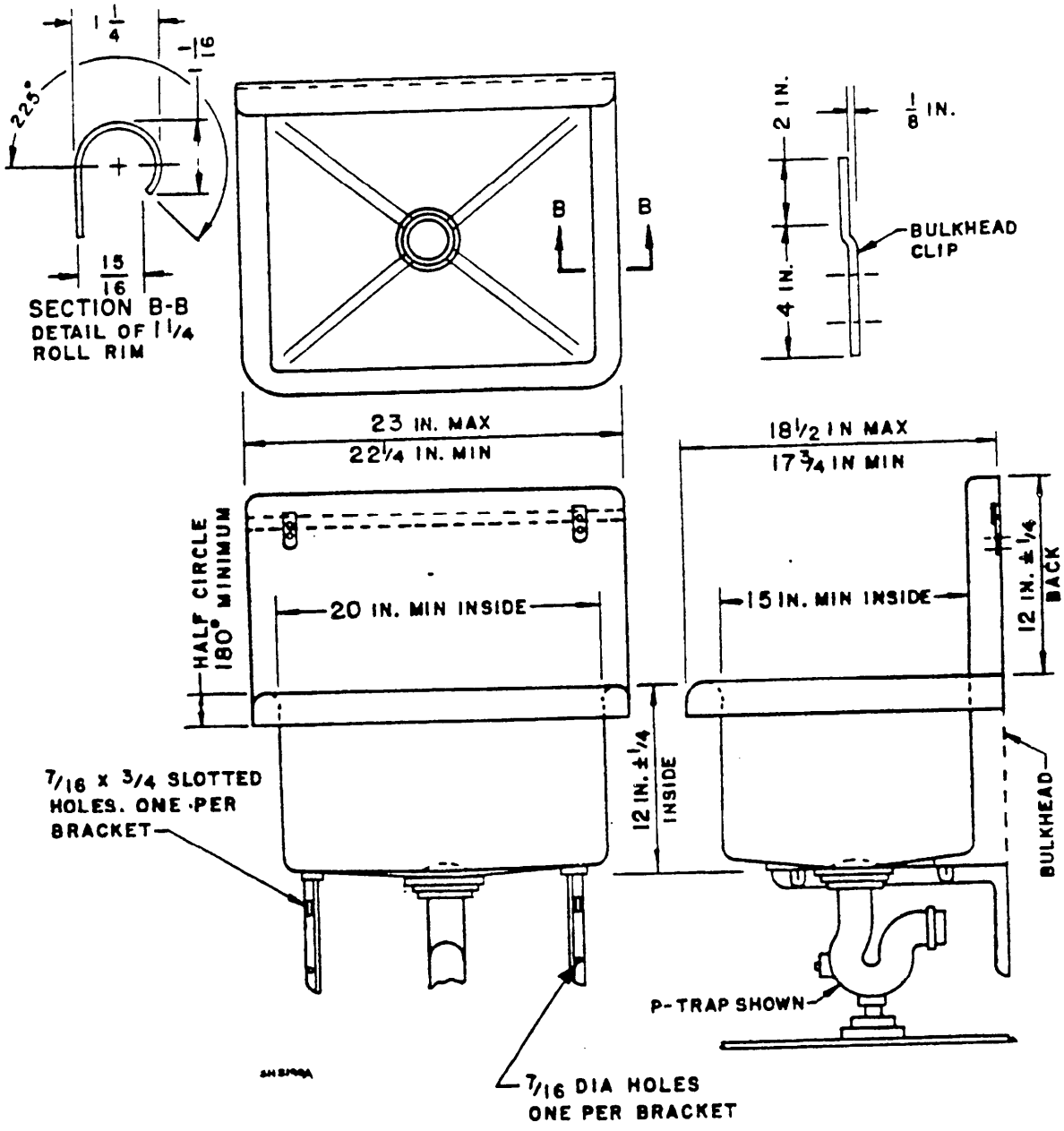



Figure 1. - Service sink.

Source: https://assist.dla.mil -- Downloaded: 20261-16T14:26Z
Check the source to verify that this is the current version before use.

TRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in pitting comments and suggestions for improvements. All users of military standardization documents are invited to provide estions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and ed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, wa rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the blems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an nowledgement will be mailed to you within 30 days to let you know that your comments were received and are being idered.

TE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of ification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document*(s)* or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

EPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATE

FICIAL BUSINESS
NALTY FOR PRIVATE USE $300

**BUSINESS REPLY MAIL**
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander, Naval Ship Engineering Center
Federal & Military Document & DOD Standardization
Branch
Department of the Navy

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

MILITARY SPECIFICATION

SINKS, SERVICE, NAVAL SHIPBOARD

This amendment forms a part of Military Specification MIL-S-2143B(SHIPS), 7 June 1967.

Page 2

3.2.3, delete and substitute:

"3.2.3 Trap. Each sink shall be fitted with a 2-inch iron pipe size (i.p.s.) cast bronze P-trap. The trap shall have a 1-inch screw clean-out plug below the water surface. Depth of water seal shall be not less than 2 inches. The trap inlet shall be flanged to provide a seating surface for the rubber gasket of the drain fitting. The trap outlet shall be a 2-inch i.p.s. silver brazed joint in accordance with MIL-F-1183. When required (see 6.1), the trap shall be provided with an adjustable support to the deck."

Page 4

6.1, add as "(e)":

"(e) Whether or not trap support is required."

Page 5

Figure 1: Delete and substitute the attached figure.

Preparing activity:
Navy-SH
(Project 4510-N0095)

FSC 4510

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.

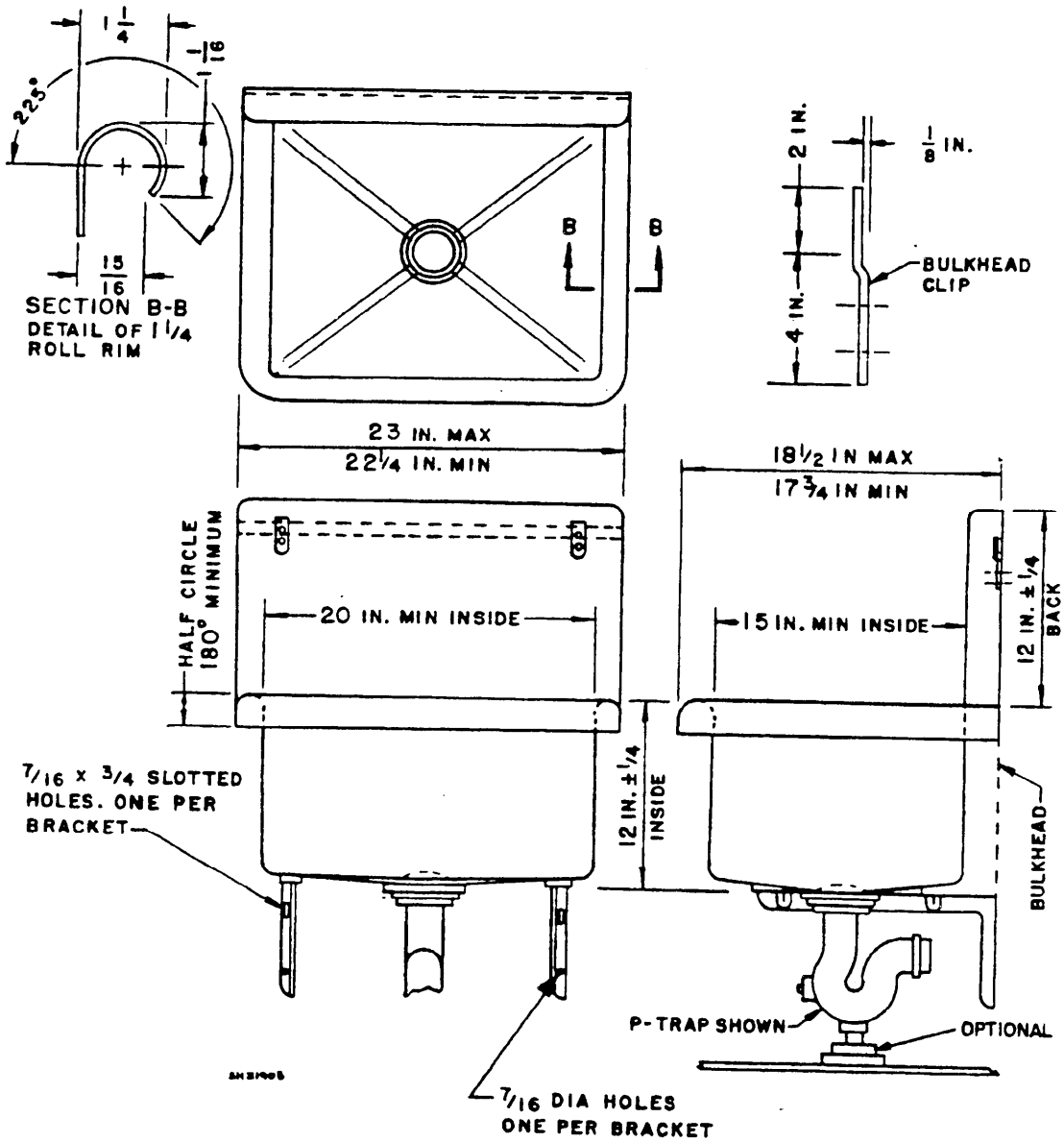



Figure 1 - Service sink.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:26Z
Check the source to verify that this is the current version before use.